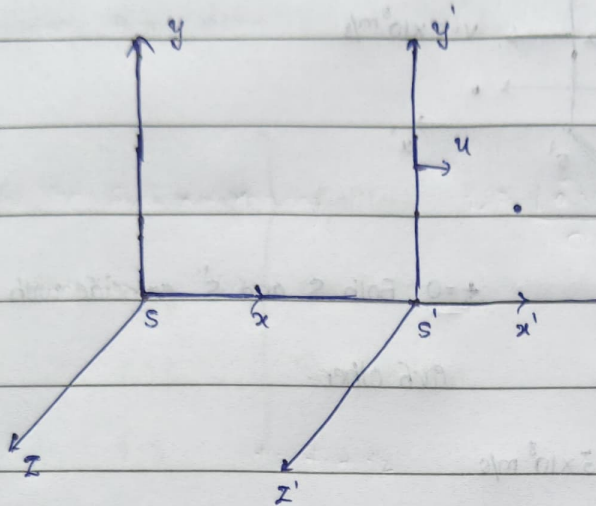


4/09/26.

Thought Exp :- Gedanken exp.

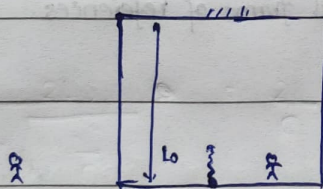


Using Galilean Transformation

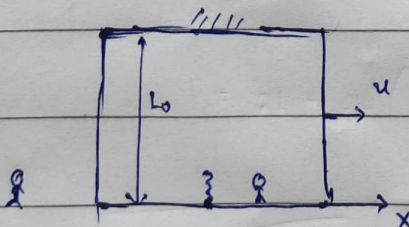
$$c' = c - u$$

Therefore length and time are not absolute. They will ~~change~~ change with the speed of the observer.

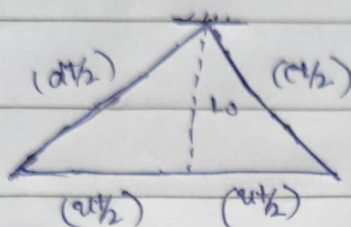
Time in a Relativistic frame :-

Case-1

$$t_0 = \frac{2L_0}{c} \quad (\text{when the car is at rest})$$

Case-2

Q: What the observer sitting outside the car will see?



$$(c(\frac{t}{2}))^2 = L_0^2 + (u(\frac{t}{2}))^2$$

$$\Rightarrow t^2 (c^2 - u^2) = 4L_0^2$$

$$t = \frac{4L_0^2}{c^2(1 - \frac{u^2}{c^2})}$$

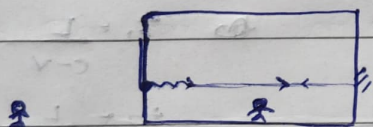
$$t = \frac{t_0}{\sqrt{1 - \frac{u^2}{c^2}}}$$

$$\text{If } u=0 \quad t=t_0$$

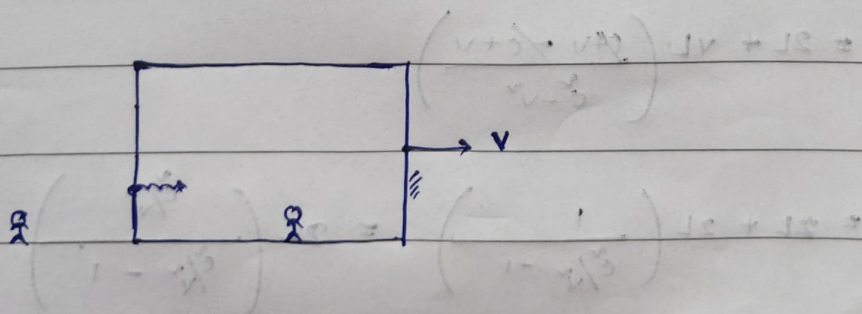
$$u \leq c \quad t \geq t_0$$

How length is changing with speed of observer?

case-1



$$t_0 = \frac{2L_0}{c} \quad (\text{when car is at rest})$$

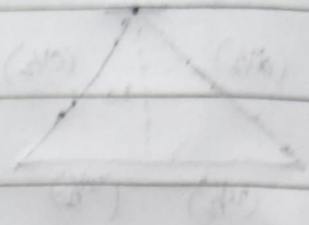


Q: What will the observer see once the car started moving relativistically?

for the forward journey

$$t_1 = \frac{L + vt_1}{c}$$

the time taken



for the backward journey

$$t_2 = \frac{L - vt_2}{c}$$

the time taken

$$t = t_1 + t_2$$

$$= \frac{L + vt_1}{c} + \frac{L - vt_2}{c}$$

$$= \frac{L}{c} + \frac{vt_1}{c} + \frac{L}{c} - \frac{vt_2}{c}$$

Exercise - Show that $L = L_0 \sqrt{1 - v^2/c^2}$

$$\frac{2L_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{2L}{c} + \frac{vL}{(c-v)c} - \frac{vL}{(c+v)c}$$

$$t_1 = \frac{L}{c-v}$$

$$t_2 = \frac{L}{c+v}$$

$$= 2L + vL \left(\frac{c+v}{c^2 - v^2} \right)$$

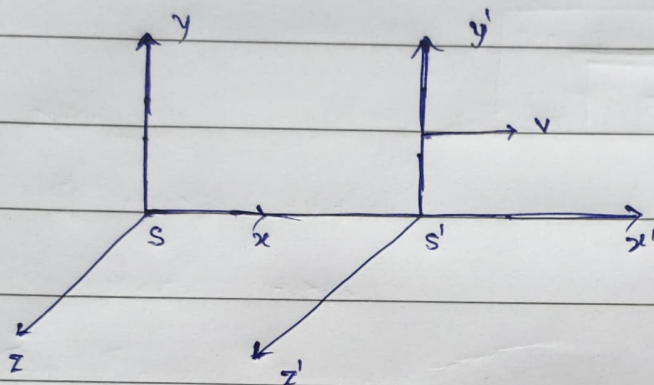
$$= 2L + 2L \left(\frac{1}{\frac{c^2}{v^2} - 1} \right) = 2L \left(\frac{\frac{c^2}{v^2}}{\frac{c^2}{v^2} - 1} \right)$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

If $v=0$ $L=L_0$

$v < c$ $L = L_0 \sqrt{1 - v^2/c^2} \Rightarrow L < L_0$

Lorentz Transformation



$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}$$

In the rest frame, the coordinates of the event is (x, y, z, t)

In the moving frame, the coordinates of the event is (x', y', z', t')

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{bmatrix} \gamma & 0 & 0 & -\gamma v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{\gamma v}{c^2} & 0 & 0 & \gamma \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}$$

$\gamma \rightarrow$ Boost factor $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$